ESSINet: Efficient Spatial–Spectral Interaction Network for Hyperspectral Image Classification

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***Abstract*— Nowadays, convolutional neural networks (CNNs) are widely used in the field of hyperspectral image (HSI) classification. However, a major feature of HSIs is their rich spectral–spatial information with hundreds of continuous bands. This inevitably incurs the problems of high computational cost for network optimization and high interredundancy in the convolution kernels. To solve these problems, in this article, we rethink HSIs from the spectral perspective and introduce a lightweight operator called involution, which can effectively solve the above limitations. Different from traditional convolution kernels, the involution kernels pay more attention to the features of the channels but usually ignore the spatial features in the receptive field. To incorporate both spatial and spectral information, we construct a dual-pooling layer and design a novel involution-2D operator and its more lightweight version, involution-1D operator. Finally, an efficient spatial–spectral interaction network (ESSINet) for HSI classification is proposed based on these two new operators, which can make the spatial–spectral information in HSIs interact more closely. Extensive experimental results on four public datasets demonstrate the effectiveness and efficiency of the proposed ESSINet over some state-of-the-art CNN-based networks.**

***Index Terms*— Efficient spatial–spectral interaction network (ESSINet), hyperspectral image (HSI) classification, lightweight network, modified involution.**

I. Introduction

HYPERSPECTRAL remote sensing technology images the ground scene at the same time through hundreds of spectral narrow bands so that the obtained HSIs contain the fine spectral information and rich spatial information of the Earth's surface materials. Hyperspectral images (HSIs) are now used widely in many fields, such as geology and minerals [1], anomaly detection [2], weather recognition [3], atmospheric sciences [4], and agriculture [5], [6].

As one of the key technologies of HSI analysis and processing, the research on HSI classification has attracted increasing attention from scholars and has become the focus of research in many fields. There are two main challenges in HSI classification tasks: 1) the large spatial variability of spectral signatures and 2) the limited and imbalanced samples versus the high dimensionality of hyperspectral data [7]. To solve them, many early HSI classification methods included a two-stage learning process. First, some feature extraction methods were used to extract discriminative information from the original HSIs. Then, classical classification models were trained based on these features. Representative works are in [8]–[11]. However, most of these methods are time-consuming and place more emphasis on feature engineering [12]. Fortunately, a series of scheduling algorithms [13], [14] were proposed, aiming to distribute the computing load on cloud computing resources, thus shortening the time required to explore HSI classification tasks to a certain extent.

Recently, deep learning (DL) methods have brought new ideas to HSI classification [15]. Specifically, Makantasis *et al.* [16] exploited a convolutional neural network (CNN) to encode the pixels' spectral and spatial information and a multilayer perceptron to conduct the classification task. Cao *et al.* [17] formulated the HSI classification problem from a Bayesian perspective and then adopted a CNN to learn the posterior class distributions. Paoletti *et al.* [18] proposed deep pyramidal residual networks (PyResNet), which increased the feature map dimension and grouped them in pyramidal bottleneck residual blocks to obtain more spatial information. However, these 2-D CNN-based methods are more inclined to extract spatial information and have a large number of parameters [19]. To reduce the number of parameters and make full use of the potential of CNNs, 3-D convolution kernels are introduced into the field of HSI classification. Lee and Kwon [20] proposed a novel deep end-to-end CNN, LeeEtAINet, which applied multiple 3-D local convolutional kernels of different sizes to jointly extract the spatial–spectral features of HSIs. Zhong *et al.* [21] proposed a spectral–spatial residual network (SSRN) to learn the discriminative features from spectral signatures and spatial contexts, and put forward the idea that batch normalization (BN) can improve the accuracy

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of HSI classification. S^3Net [22] used 3-D convolution kernels and an attention mechanism to design three branches so that the model can learn important information and suppress useless features. Roy *et al.* [23] designed a hybrid spectral convolutional neural network (HybridSN) to extract more spectral information. This model used both 2-D CNNs and 3-D CNNs to learn spatial and spectral features, respectively. However, the principal component analysis (PCA) was still used to compress the redundant spectral information, which increased the computational complexity of the model and meant that it is not an end-to-end architecture. A2S2KResNet [24], a recent work proposed by Roy *et al.*, adopted an efficient feature recalibration mechanism to boost the classification performance of HSIs. This model is not only end-to-end but also has excellent performance. However, it still requires a large amount of calculation. Although most of the 3-D CNN-based methods can reduce the number of parameters, they still have certain limitations and have high computational costs because they additionally need to traverse in depth. To solve these problems, some improved convolutional structures and other basic structures are introduced into HSI classification [25]–[27]. Different from traditional 3-D CNNs-based methods, LiteDenseNet [28], proposed by Li and Duan, used group convolution, which greatly reduced the number of parameters and computational cost. Cui *et al.* used depthwise separable convolution to construct LiteDepthwiseNet [19]. This model further reduces the number of parameters and the floating-point operations (FLOPs). Zhao *et al.* [29] presented a novel plug-and-play compact band weighting (CBW) module that made the parameters of the CNN-based models greatly reduced. On the other hand, because the correlation between spectral segments of HSIs and the correlation between video frames have certain similarities, some methods in the video field also bring inspiration to HSI classification tasks [30], [31].

It is a feasible solution to solve the inherent shortcomings of the convolution kernels and realize the lightweight of the network from the convolution structure [32]–[34]. In the field of computer vision, dynamic convolution [32] solved the disadvantage that a fixed convolution kernel can only extract a single feature and made the model more lightweight. Involution [35] combined the characteristics of dynamic convolution and attention mechanism so that it can effectively extract the features of channels and has been proven to have excellent effects in the field of computer vision. Therefore, it has great potential to be applied in HSI classification tasks. However, how to combine the spatial–spectral characteristics of HSIs to achieve effective classification is difficult. To solve this problem, we propose two novel operators, involution-1D and involution-2D, which make the involution better process the spatial–spectral information of HSIs while being lightweight and construct an efficient spatial–spectral interaction network (ESSINet) based on them. The contributions of this article are summarized as follows.

1) In view of the characteristics of hyperspectral data, a novel involution is designed, called involution-2D, where a dual-pooling layer is incorporated into the original involution to utilize the information of the spatial domain.
2) On the basis of involution-2D, the internal operators are replaced with 1-D convolution operators so that the resulted involution-1D can further reduce the FLOPs and parameters under the premise of ensuring the performance.
3) An involution-based residual block (InvBlock) is proposed to enable the spatial and spectral information to interact more closely, and a lightweight HSI classification architecture, ESSINet, is proposed based on it, which greatly reduces the number of parameters, FLOPs, inference time, and GPU memory usage.

The rest of this article is arranged as follows. In Section II, we briefly introduce the related work. The structure of the proposed ESSINet is detailed in Section III. The comparison experiment results of six algorithms on four well-known HSI datasets are reported in Section IV. Section V summarizes this article.

II. Related Work

This section starts by introducing the concept of convolution, then explains the definition of involution, and reveals the differences between them.

A. 3-D Convolution Operator and Its Properties

In the field of HSI classification, CNN-based methods have achieved excellent performance, especially the 3-D CNN-based algorithms. Compared with a 2-D convolution kernel, a 3-D convolution kernel can simultaneously capture spatial and spectral information from the original HSI. Here, a 3-D convolution is taken as an example to describe the related properties of convolution.

As shown in Fig. 1, let the input be $X \in \mathbb{R}^{C\times H\times W\times B}$, where C enumerates the channels (the number of channels of the raw HSI is 1), H and W represent the height and the width of the input, respectively, and B is the dimension of the spectral bands. $X_{c,i,j,b} \in X$ is one pixel of the input. Suppose the output $Y \in \mathbb{R}^{C_{\text{out}}\times H\times W\times B}$ is obtained after the multiply–add operation through the sliding window method, whose H, W, and B remain unchanged through padding and C_{out} represents the number of 3-D convolution kernels. Let the dimension of a 3-D convolution kernel be $K \times K \times D$, where K and D represent its size and the number of channels, respectively.

Let the kth 3-D convolution kernel be $\mathcal{F}^k \in \mathbb{R}^{C\times K\times K\times D}, k \in \{1, 2, \ldots, C_{\text{out}}\}$. $\mathcal{F}^k_{c,i,j,d} \in \mathcal{F}^k$ refers to a weight parameter in the cth channel of the kth convolution kernel. Then, a pixel of $Y_{k,i,j,b} \in Y$ can be defined as

$$Y_{k,i,j,b} = \sum_{\substack{c\in\{1,2,\ldots,C\}\\ \omega\in\Delta d\\ (u,v)\in\Delta_k}} \mathcal{F}^k_{c,u+\left[\frac{K}{2}\right],v+\left[\frac{K}{2}\right],w+\left[\frac{D}{2}\right]} X_{c,i+u,j+v,b+\omega} \quad (1)$$

where $\Delta_k = [-[K/2], \ldots, [K/2]] \times [-[K/2], \ldots, [K/2]]$, $\times$ represents the Cartesian product above, and $\Delta_d = \{-[D/2], \ldots, [D/2]\}$. Both refer to the set of offsets in the neighborhood of the convolution conducted on the center pixel.

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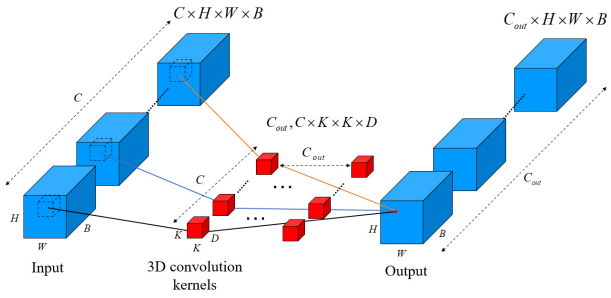

Fig. 1. 3-D convolution. The convolution performs the padding operation by default to keep the size of feature maps unchanged.

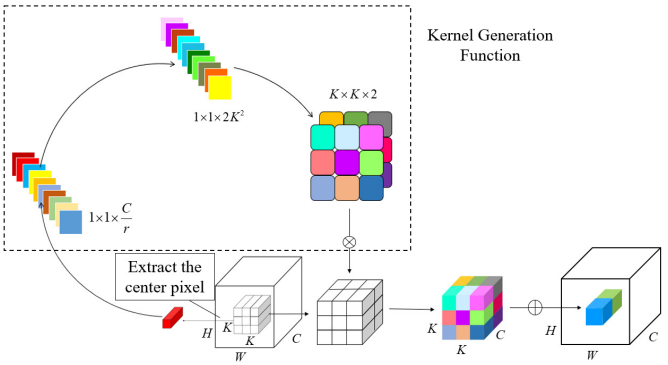

Fig. 2. Schematic of involution. For ease of understanding, the channel of the involution kernel in the figure is set to 2. $\otimes$ represents the multiplication broadcast across C channels, and $\oplus$ represents the add operation in the space.

The standard convolution kernels have two properties, namely, spatial-agnostic and channel-specific [35], [36]. The former property means that, from the spatial perspective, a single convolution kernel shares the parameters in the spatial domain, which will be repeatedly used in a feature map through the sliding window method. Channel-specific means that, from the perspective of the channel, the convolution kernel has a corresponding weight in each channel. Some works, such as [37]–[39], believe that the feature maps have redundancy in the channel dimension, and this redundancy will affect the performance of the model. However, the standard convolution kernels cannot solve it, and the problem of interchannel redundancy in convolution kernels arises [40].

B. Involution Operator and Its Properties

The concept of involution was proposed in [35]. Different from convolution kernels, the involution kernels depend on the input feature map. Specifically, let $X' \in \mathbb{R}^{H\times W\times C}$ denote the input required for involution, the size of which is also kept at $H \times W$, and the number of channels is C. $X'_{i,j,k} \in X'$ is one pixel of the input.

Let $\mathcal{H}^{i,j} \in \mathbb{R}^{K\times K\times G}$ denote the involution kernel with size $K \times K$ and the number of channels G, which is especially customized by the vector $X'_{i,j} \in \mathbb{R}^C$ located at the corresponding coordinate (i, j) in X'. Given a kernel generation function (KGF) ϕ, the involution kernel $\mathcal{H}^{i,j}$ can be defined as

$$\mathcal{H}^{i,j} = \phi(X'_{i,j}). \tag{2}$$

Denote the weight parameters in the gth channel of involution kernel as $\mathcal{H}^{i,j}_{:,:,g} \in \mathbb{R}^{K\times K}$, $g \in \{1, \ldots, G\}$. The output feature map $Y' \in \mathbb{R}^{H\times W\times C}$ is derived by performing multiply–add operations on the input with such involution kernels. Namely, $Y'_{i,j,k} \in Y'$ is obtained as follows:

$$Y'_{i,j,k} = \sum_{(u,v)\in\Delta_k} \mathcal{H}^{i,j}_{u+\left[\frac{K}{2}\right],v+\left[\frac{K}{2}\right],[\mathrm{kG}/C]} X'_{i+u,j+v,k}. \tag{3}$$

The schematic of the involution operation is shown in Fig. 2. Considering the lightweight of the model, the KGF used in [35] is composed of two layers of 1×1 convolution kernels, and therefore, it is calculated only with the central vector of the receptive field (RF). The first layer is composed of C/r convolution kernels of size $1 \times 1 \times C$, where r represents the reduction ratio, which is generally about 4. This layer is mainly for extracting features and reducing dimensionality. The second layer has GK^2 convolution kernels of size $1 \times 1 \times C/r$. BN and the rectified linear unit (ReLu) will be added between the two convolutional layers to ensure that KGF has strong feature characterization capabilities [41]. Finally, a channel-to-space rearrangement will be followed to obtain an involution kernel of size $K \times K \times G$.

In [35], the KGF ϕ is given by

$$\phi(X'_{i,j}) = W_1\sigma(W_0X'_{i,j}) \tag{4}$$

where $W_0 \in \mathbb{R}^{C/r\times C}$ and $W_1 \in \mathbb{R}^{(GK^2)\times C/r}$ imply two layers of convolution kernels, respectively, and σ represents BN and ReLu. Based on $\mathcal{H}^{i,j}$ obtained, the feature map is divided into G groups according to the channel dimension, and each group shares one channel of the involution kernel. After the multiply-add operation, we get a cube of size $1 \times 1 \times C$.

It can be found that, with different inputs, the obtained involution kernels will be different. Compared with the parameters in the convolution kernel, the parameters in the involution kernel are not shared in the spatial domain or updated through the backpropagation (BP) algorithm. In fact, it is the parameters in KGF which are shared and updated. Therefore, the involution kernel has the opposite properties, namely, spatial-specific and channel-agnostic. Spatial-specific means that the involution kernel adjusts the weight parameters according to the different spatial positions to extract the discriminative features. Channel-agnostic means that the involution kernel shares the parameters along the channel dimension [35].

III. Proposed ESSINet

In this section, we describe modified involution operators and the proposed ESSINet in detail.

A. Involution-2D

The original KGF only uses a vector in the middle of the RF to generate an involution kernel, that is, it tries to generate a kernel only using the spectral information while ignoring the spatial features of the RF. Therefore, we optimize involution from the perspective of modifying KGF.

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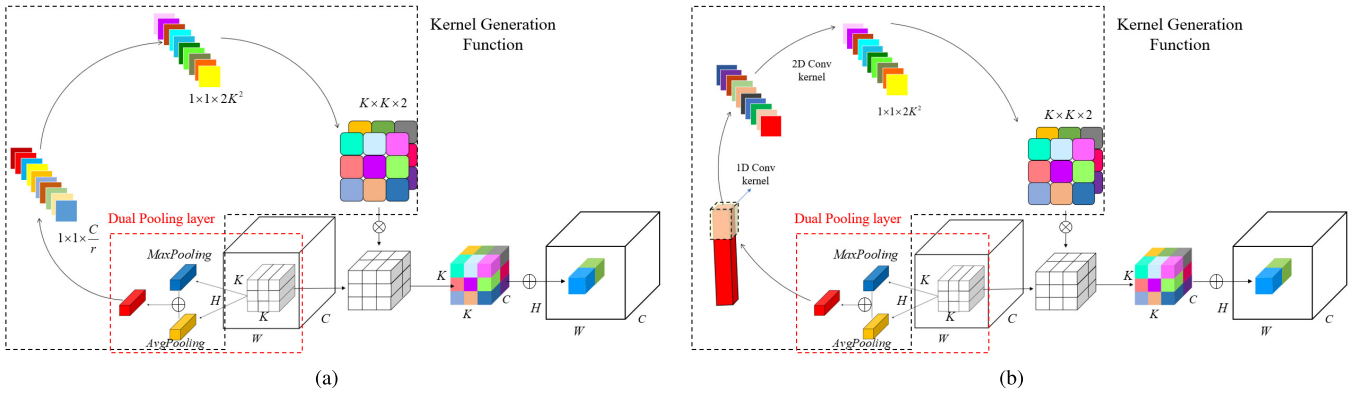

Fig. 3. Schematic of modified involution operators. Both modified involution operators need to integrate spatial features through the dual-pooling layer. The difference is that the left involution-2-D is composed of 2-D convolution operators, while the right involution-1D replaces the 2-D convolution kernels with 1-D convolution kernels. (a) Involution-2D. (b) Involution-1D.

The traditional pooling layer can integrate the features in each area of the feature map to obtain a new feature map. This technology not only achieves the purpose of reducing the number of parameters and calculations; it also strengthens the feature expression ability of the model [42]. According to this idea of the pooling operation, we apply it to the KGF of involution and propose a dual-pooling layer. As shown in Fig. 3(a), different from the original method that only extracts the center vector of the RF as an input of KGF, we enhance the feature expression ability of KGF using the dual-pooling layer. In more detail, the dual-pooling layer is composed of a max-pooling layer (MaxPooling) and an average pooling layer (AvgPooling). We use these two layers on RF to extract as many features as possible. For the current RF, two cubes of size $1 \times 1 \times C$ can be simultaneously obtained by the dual-pooling layer and then integrated to obtain a new cube of size $1 \times 1 \times C$. Compared with the original input, this new cube integrates most of the information in the corresponding RF. In addition, for the points on the edge of the feature map, we perform a padding operation so that the feature map copies the values from the border area to the padding area. Based on the above description, the modified KGF ϕ_{new} can be written as follows:

$$\phi_{\text{new}}(\Psi_{i,j}) = W_1\sigma(W_0 D(\Psi_{i,j})) \tag{5}$$

where $D(\cdot)$ represents the dual-pooling layer and $\Psi_{i,j}$ represents the RF centered on coordinate (i, j).

All in all, the function of the dual-pooling layer is to integrate the information of the spatial scope as much as possible and form a complementary relationship with the original KGF. This modified KGF enables involution to have stronger feature extraction capabilities without introducing more parameters.

B. Involution-1D

There are many articles devoted to proposing lightweight HSI classification models, such as [19], [28], [29]. In this article, we improve the involution again and propose a more lightweight structure, called involution-1D. We notice that all the parameters of the involution are generated in KGF, which is realized with the help of the 2-D convolution kernels. One of the characteristics of the 2-D convolution kernel is that its channel dimension must be consistent with the channel dimension of the feature maps, which makes the number of parameters of the 2-D convolution kernels relatively large, even if we only use 1×1 convolution kernels. Inspired by [29], we replace the 2-D convolution kernels in the involution with 1-D convolution kernels. This change can greatly reduce the number of parameters in KGF. Its specific structure is shown in Fig. 3(b).

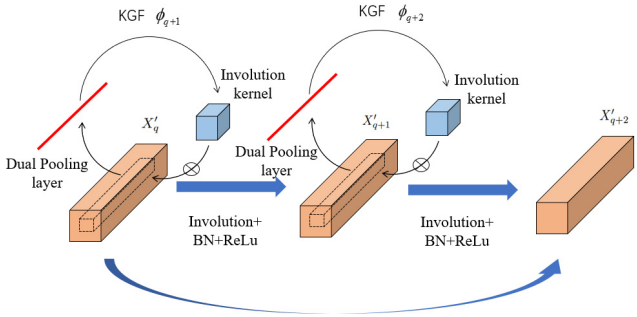

Fig. 4. Involution-based residual block.

C. Involution-Based Residual Block and ESSINet

The residual connection has been proven to be effective in HSI classification [21] and can prevent the problem of gradient vanishing or exploding [43]. Therefore, in this article, we construct the involution-based blocks through the residual connection and get the proposed ESSINet by stacking this InvBlock.

As shown in Fig. 4, ϕ_q and ϕ_{q+1} represent the KGF of the qth layer and the $(q+1)$th layer, respectively. X'_q represents the feature map of the qth layer. With the sliding window method, the output feature maps X'_{q+1} and X'_{q+2} can be obtained, respectively, after the involution kernel performs the multiply–add operation with the input feature map. Furthermore, we also combine BN and ReLu to construct an InvBlock because the combination of them and the residual

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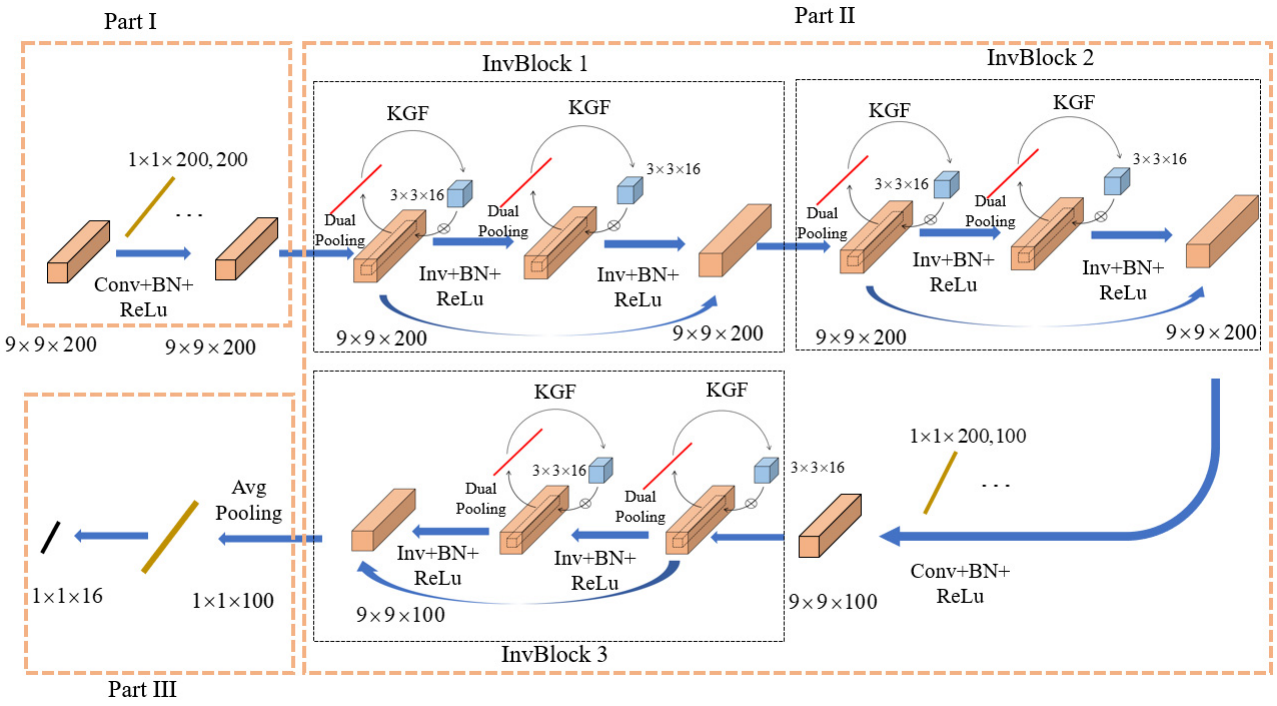

Fig. 5. Framework of ESSINet. Take a patch with the size of $9 \times 9 \times 200$ as an example. The model comprises three parts. Part I is the feature extraction, Part II is the space-spectral interaction, which is mainly composed of the modified involution, and Part III is the classification.

connection have enabled practitioners to train significantly better networks [41]. Finally, these two involutional layers need to build a residual function $F(X'_q)$. The architecture can be formulated as follows:

$$X'_{q+2} = X'_q + F(X'_q). \tag{6}$$

Here, the models with the two proposed involution operators are named ESSINet-2D and ESSINet-1D, respectively. The ESSINet-2D can be divided into three parts, and its structure diagram is shown in Fig. 5. For the convenience of explanation, we use a patch of size $9 \times 9 \times 200$ to illustrate our proposed model.

In Part I, for the input of size $9 \times 9 \times 200$, we extract the features from the spectral dimension through 200 2-D convolution kernels of size $1 \times 1 \times 200$ and obtain the output of size $9 \times 9 \times 200$. The 1×1 convolution kernels are used here mainly to integrate the information of channels in the raw HSI rather than to reduce dimensionality. Part II is the core module of the model. For the output of Part I, it will enter the InvBlock twice in a row to make the spatial and spectral information fully interact, and then, an output of the same size will be obtained. Part II also uses 100 convolution kernels of size $1 \times 1 \times 200$. Its purpose is to reduce the dimension of the feature maps and prevent the full connection (FC) layer from having too many parameters, which may lead to overfitting [21]. Then, its output feature map of size $1 \times 1 \times 200$ is transferred to InvBlock again to further allow the spatial and spectral information in the feature map to interact. In general, in this part, the model continuously integrates spatial–spectral features and promotes the interaction between them in the process of information dissemination to obtain better involution kernels for the purpose of extracting discriminative features. Ultimately, the final result is obtained through a global average pooling layer and a fully connected layer in Part III. It is worth mentioning that the model adds 50% dropout to the FC layer to prevent overfitting.

Considering the lightweight of the model and the characteristics of the proposed involution-1D operator, the ESSINet-1D only includes Part II and Part III of Fig. 5. The main difference is that the involution-2D operator of the InvBlock is replaced by the involution-1D operator.

D. Comparing Calculation Cost and Number of Parameters

Compared with standard 2-D convolution, the involution has fewer parameters and calculational consumption. Suppose there is a feature map of size $H \times W \times C$; then, the padding operation will be automatically performed so that the size of the feature maps remains unchanged. For a convolution kernel of size $K \times K$, to obtain an output of size $Y \in \mathbb{R}^{H \times W \times C}$, the required number of parameters is

$$K \times K \times C \times C = C^2K^2 \tag{7}$$

and the computation cost is

$$K \times K \times C \times C \times H \times W = \mathrm{HWC}^2K^2. \tag{8}$$

However, for an involution-2D kernel of size $K \times K$, the number of parameters is

$$C \times \frac{C}{r} + \frac{C}{r} \times G \times K \times K = \frac{\left(C^2 + \mathrm{CG}K^2\right)}{r} \tag{9}$$

and the calculation cost is

$$\mathrm{HW} \times \left(\frac{K^2\mathrm{GC} + C^2}{r} + K^2C\right). \tag{10}$$

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For involution-1D, assuming that the size of a 1-D convolution kernel in KGF is T and the stride is s, then the number of parameters is

$$T + \frac{C - T + s}{s} \times G \times K \times K \tag{11}$$

and the calculation cost is

$$\mathrm{HW} \times \left(T + \frac{C - T + s}{s} \times G \times K \times K + K^2 C \right). \tag{12}$$

As for the 3-D convolution kernel, since it needs to traverse in depth, the computational complexity of 3-D CNN-based methods tends to be higher.

IV. Experiment Results

We utilize four widely used HSI datasets to compare ESSINet-2D and ESSINet-1D with the other six models, namely, PyResNet [18], CBW [29], HybridSN [23], LeeEtAlNet [20], SSRN [21], and A2S2KResNet [24]. According to their basic structure, these six models can be divided into 2-D CNN-based models, 1-D-2-D CNN-based models, 2-D-3-D CNN-based models, and 3-D CNN-based models, respectively. Three quantitative metrics, overall accuracy (OA), average accuracy (AA), and κ coefficient (Kappa), are used to evaluate the performance of each method. We also use four metrics to evaluate the models' efficiency: FLOPs, the number of parameters, inference time (s), and GPU memory usage (MB).

All the experiments are implemented on Ubuntu 18.04.5 long term support (LTS), CPU: i9 9900k, GPU: 2080Ti, random access memory (RAM): 64 GB, GPU memory: 11 GB, Python: 3.8.5, and Torch: 1.9.

A. Data Description

The following four open HSI datasets are used in the experiments of this article, namely, Indian Pines (IP), Pavia University (PU), Salinas scene (SA), and the University of Houston campus (HS).[1]

1) Indian Pines (IP): This dataset was captured by sensors located in Northwest Indiana by the Airborne Visible Infrared Imaging Spectrometer (AVIRIS). The size of the image is 145×145, containing 200 spectral bands, and the spatial resolution is 20 m per pixel. The wavelength ranges from 0.4 to 2.5 μm. Its ground truth (GT) can be divided into 16 types of categories.

2) Pavia University (PU): This dataset was acquired in 2001 by the reflector optical imaging spectroradiometer in the area around the PU in Italy from sensors in northern Italy. The image size is 610×340, containing 103 spectral reflection bands, the spatial resolution is 1.3 m per pixel, and the wavelength ranges from 0.43 to 0.86 μm. This image contains nine different categories.

3) Salinas Scene (SA): This dataset was collected by a 224-band AVIRIS sensor over the Salinas Valley in California, with a high spatial resolution (3.7 m per pixel). The image size is 512×217 and contains 204 spectral reflection bands. This image contains 16 different classes.

[1]The IP, PU, and SA datasets can be obtained from http://www.ehu.eus/ccwintco/index.php?title=Hyperspectral_Remote_Sensing_Scenes, and the HS dataset can be obtained from https://hyperspectral.ee.uh.edu/?page_id=459

TABLE I
Samples for Each Category of Training, Validation, and Testing for the IP Dataset

No.	Class	Training	Validation	Testing	Total Number
1	Alfal	5	5	36	46
2	Corn-notill	143	143	1142	1428
3	Corn-mintill	83	83	664	830
4	Corn	24	24	189	237
5	Grass-pasture	48	48	387	483
6	Grass-trees	73	73	584	730
7	Grass-pasture-mowed	6	3	19	28
8	Hay-windrowed	48	48	382	478
9	Oats	4	2	14	20
10	Soybean-notill	97	97	778	972
11	Soybean-mintill	246	246	1964	2455
12	Soybean-clean	59	59	474	593
13	Wheat	21	21	163	205
14	Woods	127	127	1012	1265
15	Buildings-Grass-Trees	39	39	308	386
16	Stone-Steel-Towers	9	9	75	93
	Total	1032	1027	8190	10249

TABLE II
Samples for Each Category of Training, Validation, and Testing for the HS Dataset

No.	Class	Training	Validation	Testing	Total Number
1	Healthy grass	63	63	1125	1251
2	Stressed grass	63	63	1128	1254
3	Synthetic grass	35	35	627	697
4	Trees	62	62	1120	1244
5	Soil	62	62	1118	1242
6	Water	16	16	293	325
7	Residential	63	63	1142	1268
8	Commercial	62	62	1120	1244
9	Road	63	63	1126	1252
10	Highway	61	61	1105	1227
11	Railway	62	62	1111	1235
12	Parking Lot 1	62	62	1109	1233
13	Parking Lot 2	23	23	423	469
14	Lettuce romaine 7wk	21	21	386	428
15	Tennis Court	33	33	594	660
	Total	751	751	13527	15029

4) University of Houston Campus (HS): This dataset is provided by the HSI Analysis Group and the Airborne Laser Mapping Center (NCALM) funded by the University of Houston. It has a high spatial resolution (2.5 m) and has 144 spectral bands from 380 to 1050 nm. The image size is 349×1905. There are 15 categories in this dataset.

Due to the serious category imbalance problem in the IP dataset, we take 10% of the samples on the IP dataset as the training set. In the follow-up experiments, 5%, 1%, and 1% of the data of the HS, SA, and PU datasets will be used as a training set, respectively. The ratio of the validation set is the same as the training set, and the rest data are used as the testing set. The detailed information of the four datasets is listed in Tables I–IV, respectively.

B. Framework Setting

First, the focal loss [44] function in the proposed ESSINet is defined as

$$\mathrm{FL}(p, \alpha, \gamma, y) = -F_y(\alpha)(1 - F_y(\alpha))^{\gamma} \log(F_y(\alpha)) \tag{13}$$

where $\alpha \in [0, 1]$, $\gamma \geq 0$ and

$$F_y(\alpha) = \begin{cases} \alpha, & y = 1 \\ 1 - \alpha, & \text{otherwise.} \end{cases} \tag{14}$$

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TABLE III
SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION, AND TESTING FOR THE SA DATASET

No.	Class	Training	Validation	Testing	Total Number
1	Broccoli green weeds 1	20	20	1969	2009
2	Broccoli green weeds 2	37	37	3652	3726
3	Fallow	20	20	1936	1976
4	Fallow rough plow	14	14	1369	1397
5	Fallow-smooth	27	27	2624	2678
6	Stubble	40	40	3879	3959
7	Celery	36	36	3507	3579
8	Grapes untrained	113	113	11045	11271
9	Soil vinyard develop	62	62	6079	6203
10	Corn senesced green weeds	33	33	3212	3278
11	Lettuce romaine 4wk	11	11	1046	1068
12	Lettuce romaine 5wk	20	20	1887	1927
13	Lettuce romaine 6wk	9	9	898	916
14	Lettuce romaine 7wk	11	11	85	1070
15	Vinyard untrained	73	73	7122	7268
16	Vinyard vertical trellis	18	18	1771	1807
	Total	544	544	53041	54129

TABLE IV
SAMPLES FOR EACH CATEGORY OF TRAINING, VALIDATION, AND TESTING FOR THE PU DATASET

No.	Class	Training	Validation	Testing	Total Number
1	Asphalt	66	66	6499	6631
2	Meadows	187	187	18275	18649
3	Gravel	21	21	2057	2099
4	Trees	31	31	3002	3064
5	Painted metal sheets	14	14	1317	1345
6	Bare Soil	50	50	4929	5029
7	Bitumen	13	13	1304	1330
8	Self-Blocking Bricks	37	37	3608	3682
9	Shadows	10	10	927	947
	Total	429	429	41918	42776

The focal loss function can focus on both small-category samples and difficult-to-classify samples, so it is very suitable for HSI datasets with sample imbalance problems [7]. Then, we use RMSprop as the optimizer and use the linear warm-up strategy, a strategy of using less aggressive learning rates at the start of training. It first linearly increases the learning rate from zero to the specified learning rate and then linearly reduces the learning rate, which can make the model converge healthily [45]. Its warm-up step is 100. Finally, the learning rate of the PU dataset is set to 0.005, and the other datasets are set to 0.003 according to the results of the grid search method.

For all the experiments, we set the focal loss γ to 2, and α is the inverse of the number of samples in each category. The models have been trained for 500 epochs, and the batch size of all experiments is 128.

C. Classification Maps and Categorized Results

1) Classification Maps and Categorized Results for the IP Dataset: The results of different algorithms to classify the IP dataset are shown in Table V, and the classification maps of different algorithms are shown in Fig. 6. The main problem with this dataset is that the data distribution is very imbalanced, so it is difficult to classify some categories. It can be seen that ESSINet-2D performs best compared with the other six algorithms in this dataset, and the classification map of ESSINet-2D has clear boundaries and less noise. From the perspective of model efficiency, ESSINet-1D has the least number of parameters and FLOPs, and although its performance has dropped, it is still comparable to 3-D CNN-based models. In most categories, A2S2KResNet achieves similar results to SSRN, and these two methods also achieve satisfactory results on the IP dataset, second only to ESSINet. Compared with 3-D CNN-based models, the 2-D-3-D CNN-based models have fewer FLOPs, but their classification abilities are not as good as theirs. It is worth noting that the overall performance of HybridSN is not satisfactory. For PyResNet and CBW, CBW achieves better classification results on the basis of greatly reducing the number of parameters and FLOPs, but it is difficult to achieve results similar to 3-D CNN-based models and ESSINets.

2) Classification Maps and Categorized Results for the HS Dataset: Table VI and Fig. 7 show the classification results and maps of different models on the HS dataset, respectively. The HS dataset has a balanced distribution of categories and a large number of data. ESSINet-2D achieves the best results on OA, AA, and Kappa. It is nearly 1% better than SSRN on OA and Kappa, and improved by about 0.5% on AA. Among the comparison models, the performance of SSRN is second only to our model, and LeeEtAINet in the 2-D-3-D CNN-based models does not perform well on this dataset. PyResNet and CBW perform better than the 2-D-3-D CNN-based models. In terms of the model efficiency, ESSINet-1D's FLOPs on HS is only 3M, which is much smaller than the comparison models. CBW, which is the closest to ESSINet-1D's FLOPs in comparison models, has FLOPs of 10M, but its accuracy is lower than ESSINet-1D. For the other three metrics, ESSINets also achieve the best results.

3) Classification Maps and Categorized Results for the SA Dataset: Table VII and Fig. 8 show the classification results and maps of the different models on the SA dataset, respectively. The proposed algorithms achieve satisfactory results with only 1% of the data. It can be seen that ESSINet-2D's OA, AA, and Kappa reach 97.895%, 98.771%, and 97.655%, respectively, which is about 0.7%, 0.06%, and 0.6% better than SSRN and about 1.6%, 2.1%, and 0.3% better than A2S2KResNet. From the perspective of classification maps, its classification map has clearer boundaries and less noise in the details. The two 3-D CNN-based models, SSRN and A2S2KResNet, also have good results on this dataset, but they have high FLOPs in the comparison models. Meanwhile, HybridSN and LeeEtAINet do not perform well on the SA dataset in the case of a small number of samples. In terms of model efficiency, ESSINet-1D achieves great OA, AA, and Kappa comparable to 3-D CNN models with the least number of parameters and FLOPs, and its GPU memory usage and inference time are also very low. By contrast, although SSRN and A2S2KResNet also perform well, they are time-consuming. It is worth mentioning that CBW has the smallest GPU memory usage and is second only to our models in FLOPs and the number of parameters, but its performance is not good with a small number of training samples.

4) Classification Maps and Categorized Results for the PU Dataset: Table VIII presents the quantitative results of the

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TABLE V
CATEGORIZED RESULTS FOR THE IP DATASET USING 10% TRAINING SAMPLES

Class	Color	2D CNNs	1D-2D CNNs	2D-3D CNNs		3D CNNs		Involution	
		PyResNet	CBW	HybridSN	LeeEtAlNet	A2S2KResNet	SSRN	ESSINet-2D	ESSINet-1D
1		75.00	75.676	37.5	65.625	**96.875**	**96.875**	**96.875**	89.190
2		90.242	90.989	72.366	81.347	**99.309**	98.705	98.791	98.425
3		84.101	90.512	72.214	95.691	99.108	99.554	**99.703**	97.892
4		91.813	83.684	90.058	100.00	**99.415**	97.661	**99.415**	98.421
5		95.153	92.746	97.194	96.429	97.704	97.704	**98.214**	94.041
6		97.635	99.829	99.155	98.142	99.493	99.324	98.986	**99.658**
7		94.444	90.909	77.778	100.00	94.444	94.444	**100.00**	**100.00**
8		98.969	**100.00**	**100.00**	99.742	**100.00**	**100.00**	**100.00**	**100.00**
9		81.250	**100.00**	0	81.818	**100.00**	72.727	**100.00**	93.750
10		91.117	91.131	79.442	94.924	96.447	98.096	97.208	**98.715**
11		91.654	95.418	88.537	92.810	98.793	98.341	**99.296**	98.116
12		78.170	83.158	86.902	87.526	**98.545**	96.050	97.921	98.316
13		99.324	**100.00**	**100.00**	**100.00**	**100.00**	99.324	**100.00**	**100.00**
14		96.491	99.802	97.271	98.441	99.805	**100.00**	99.805	99.506
15		98.566	86.408	96.057	98.425	**100.00**	**100.00**	**100.00**	99.029
16		97.143	98.609	98.571	**100.00**	**100.00**	**100.00**	95.714	95.945
OA ×100		91.906	93.609	87.001	93.354	98.904	98.685	**99.002**	98.402
AA ×100		91.921	91.260	80.815	93.214	98.746	96.800	**98.871**	97.563
Kappa ×100		90.757	92.694	85.104	92.416	98.748	98.498	**98.859**	98.180
FLOPs		42M	11M	60M	32M	158M	171M	15M	**4M**
Parameters		21.911M	1.603M	3.536M	554.36k	370.86k	364.17k	134.86k	**42.90k**
Inference Time		1.87s	1.06s	3.01s	3.96s	3.94s	3.00s	**1.00s**	1.13s
GPU Memory Usage		1347M	**1145M**	1347M	1237M	2111M	1645M	1251M	1237M

TABLE VI
CATEGORIZED RESULTS FOR THE HS DATASET USING 5% TRAINING SAMPLES

Class	Color	2D CNNs	1D-2D CNNs	2D-3D CNNs		3D CNNs		Involution	
		PyResNet	CBW	HybridSN	LeeEtAlNet	A2S2KResNet	SSRN	ESSINet-2D	ESSINet-1D
1		95.826	96.092	93.250	91.350	97.957	98.046	**99.301**	97.780
2		99.203	99.469	98.760	96.722	99.469	99.646	**99.900**	99.646
3		99.202	99.681	99.841	98.883	**99.841**	99.840	99.821	99.681
4		97.679	97.768	99.482	85.625	**100.00**	**100.00**	95.880	98.036
5		98.390	100.00	97.227	96.243	**100.00**	**100.00**	99.598	99.911
6		86.391	91.438	90.753	81.264	97.603	**97.945**	93.856	**97.945**
7		88.957	95.793	84.312	83.260	97.020	98.159	98.229	**98.247**
8		85.893	89.911	80.625	78.125	93.929	89.821	**95.879**	90.893
9		85.803	82.697	82.786	78.438	85.093	87.400	**94.212**	90.861
10		92.844	95.199	92.210	93.659	96.659	97.917	**99.593**	99.457
11		95.054	90.737	84.982	86.691	96.043	**99.191**	99.190	95.235
12		92.252	94.685	88.559	91.171	96.667	96.757	**98.479**	98.288
13		83.412	93.128	91.232	39.336	97.630	**97.867**	94.478	94.076
14		98.961	**100.00**	96.623	94.545	**100.00**	**100.00**	**100.00**	99.221
15		99.495	**100.00**	**100.00**	99.158	**100.00**	**100.00**	**100.00**	**100.00**
OA ×100		93.450	94.810	91.232	87.610	96.791	97.139	**98.062**	97.109
AA ×100		93.285	95.107	91.976	86.294	97.200	97.506	**97.924**	96.875
Kappa ×100		92.920	94.389	90.526	86.595	96.531	96.907	**97.904**	97.285
FLOPs		38M	10M	42M	26M	122M	113M	9M	**3M**
Parameters		21.735M	1.598M	2.505M	472.08k	283.44 k	278.13k	79.86k	**28.98k**
Inference Time		1.59s	1.01s	3.27s	3.11s	2.86s	3.29s	**0.95s**	0.96s
GPU Memory Usage		1339M	**1195M**	1257M	1235M	1837M	1507M	1229M	1223M

different methods on the PU dataset, and Fig. 9 shows the classification maps of the different algorithms and the GT. In the case of using only 1% of the training samples, SSRN, A2S2KResNet, and ESSINets are still better than the other algorithms, each of which exceeds 97%, while ESSINet-2-D has the best performance, reaching 98.736%, 97.760%, and 98.322% on OA, AA, and Kappa, respectively. It can be clearly seen that 2-D-3-D CNN-based methods do not perform well at a small training rate, such as LeeAtAINet, which only achieves an OA of 76.892%. CBW still outperforms PyResNet. On the other hand, a comparison of model efficiency shows that the FLOPs of ESSINet-1D on this dataset are nearly

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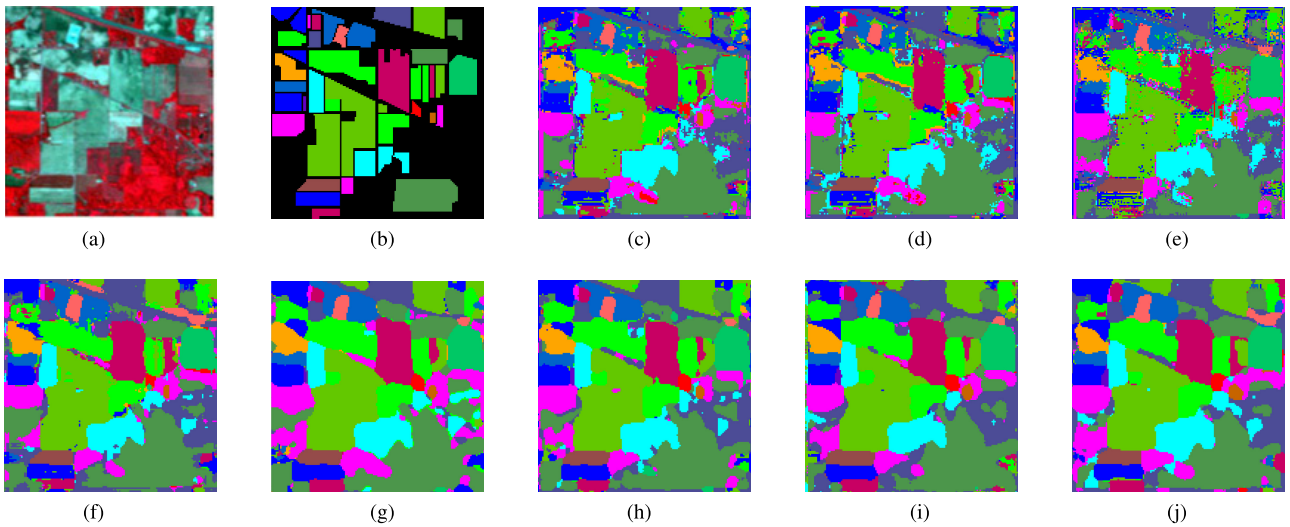

Fig. 6. Classification maps for the IP dataset using 10% of the training samples. (a) False-color image. (b) GT. (c)–(j) Classification maps with disparate algorithms.

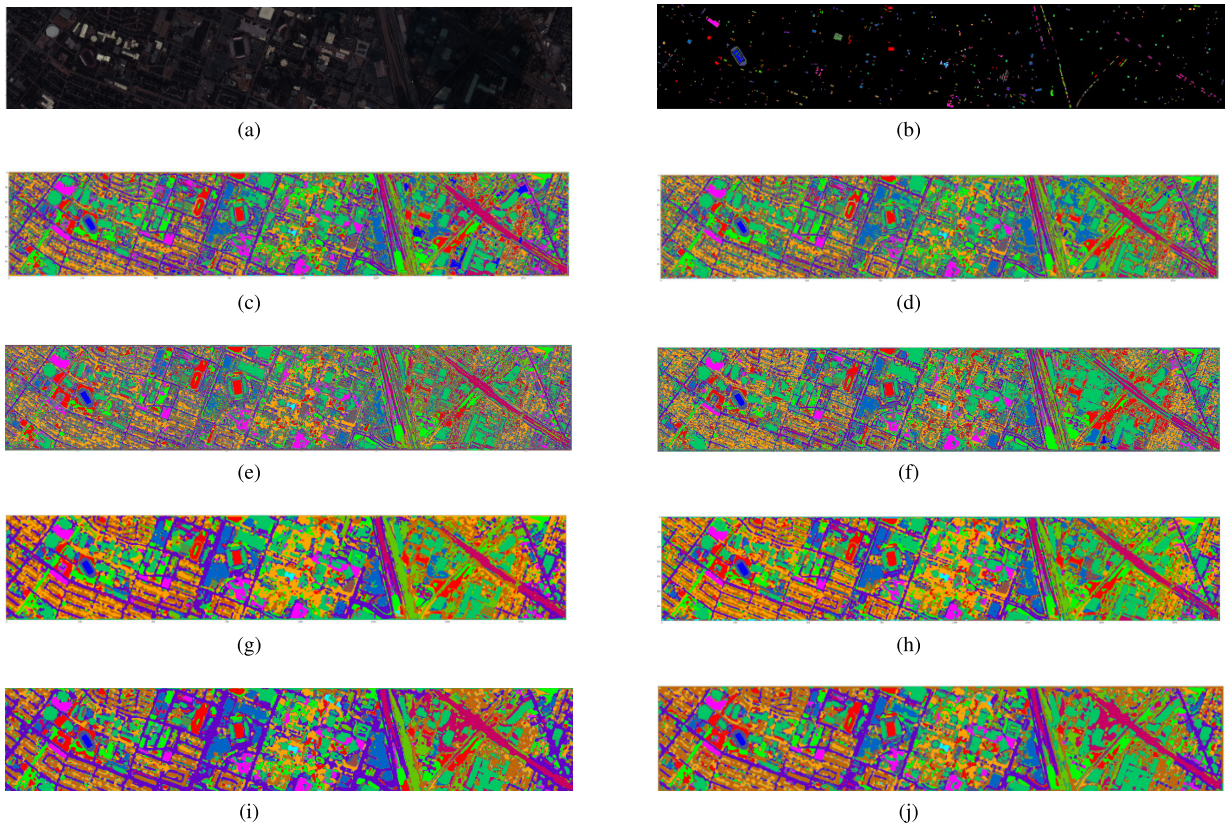

Fig. 7. Classification maps for the HS dataset using 5% of the training samples. (a) False-color image. (b) GT. (c)–(j) Classification maps with different algorithms.

88 times smaller than that of A2S2KResNet, and the number of parameters is 11 times smaller.

D. Analysis of Model Efficiency

Fig. 10 and Tables V–VIII compare the number of parameters, FLOPs, inference time, and GPU memory usage for the different models under different datasets. In general, it can be found that the proposed ESSINets achieve good results on all four metrics.

1) On **FLOPs**, the FLOPs of ESSINets do not exceed 15M on these four datasets, especially ESSINet-1D, which is nearly three times smaller than ESSINet-2D. Especially on the PU dataset, ESSINet-1D only needs 1M FLOPs, which is 80 times less than that of 3-D CNN-based models. The 3-D CNN-based models have the largest

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TABLE VII
CATEGORIZED RESULTS FOR THE SA DATASET USING 1% TRAINING SAMPLES

Class	Color	2D CNNs	1D-2D CNNs	2D-3D CNNs		3D CNNs		Involution	
		PyResNet	CBW	HybridSN	LeeEtAlNet	A2S2KResNet	SSRN	ESSINET-2D	ESSINET-1D
1		99.695	100.00	98.222	97.613	**100.00**	**100.00**	**100.00**	**100.00**
2		99.315	100.00	98.576	88.007	**100.00**	**100.00**	99.945	99.973
3		99.019	97.417	99.563	99.189	**99.973**	99.945	99.948	99.948
4		96.193	100.00	98.902	97.291	**99.854**	99.634	99.634	98.902
5		99.542	97.447	97.256	95.198	99.619	99.314	99.314	**100.00**
6		98.273	100.00	98.711	96.521	99.381	**100.00**	**100.00**	97.912
7		99.772	100.00	99.230	88.024	**100.00**	**100.00**	**100.00**	**100.00**
8		88.122	83.940	84.582	75.140	89.616	91.517	91.517	**94.695**
9		99.177	99.869	97.927	93.052	98.737	99.720	99.720	**99.901**
10		94.177	96.949	91.283	88.356	99.688	**100.00**	98.163	97.229
11		91.881	95.511	93.601	91.885	96.275	**98.376**	**98.376**	96.371
12		99.894	98.994	96.081	88.718	**100.00**	**100.00**	**100.00**	**100.00**
13		99.666	99.109	99.443	93.541	99.889	**100.00**	**100.00**	97.884
14		95.996	96.568	88.560	53.194	**98.570**	98.189	98.189	97.331
15		78.028	68.847	52.787	57.167	94.595	**94.960**	**94.960**	90.510
16		98.701	99.943	93.281	57.369	99.774	99.774	**100.00**	99.774
OA ×100		93.515	91.843	87.269	82.046	96.304	97.296	**97.895**	97.089
AA ×100		96.126	95.912	90.945	84.665	96.676	98.718	**98.771**	98.152
Kappa ×100		92.777	90.912	85.778	79.970	97.342	96.993	**97.655**	96.758
FLOPs		42M	11M	61M	32M	175M	161M	15M	**4M**
Parameters		21.923M	1.603M	3.611M	559.25k	377.1k	370.31k	138.97k	**44.41k**
Inference Time		5.00s	3.22s	8.15s	18.79s	19.91s	15.06s	**3.03s**	4.03s
GPU Memory Usage		1345M	**1197M**	1295M	1237M	2115M	1645M	1267M	1261M

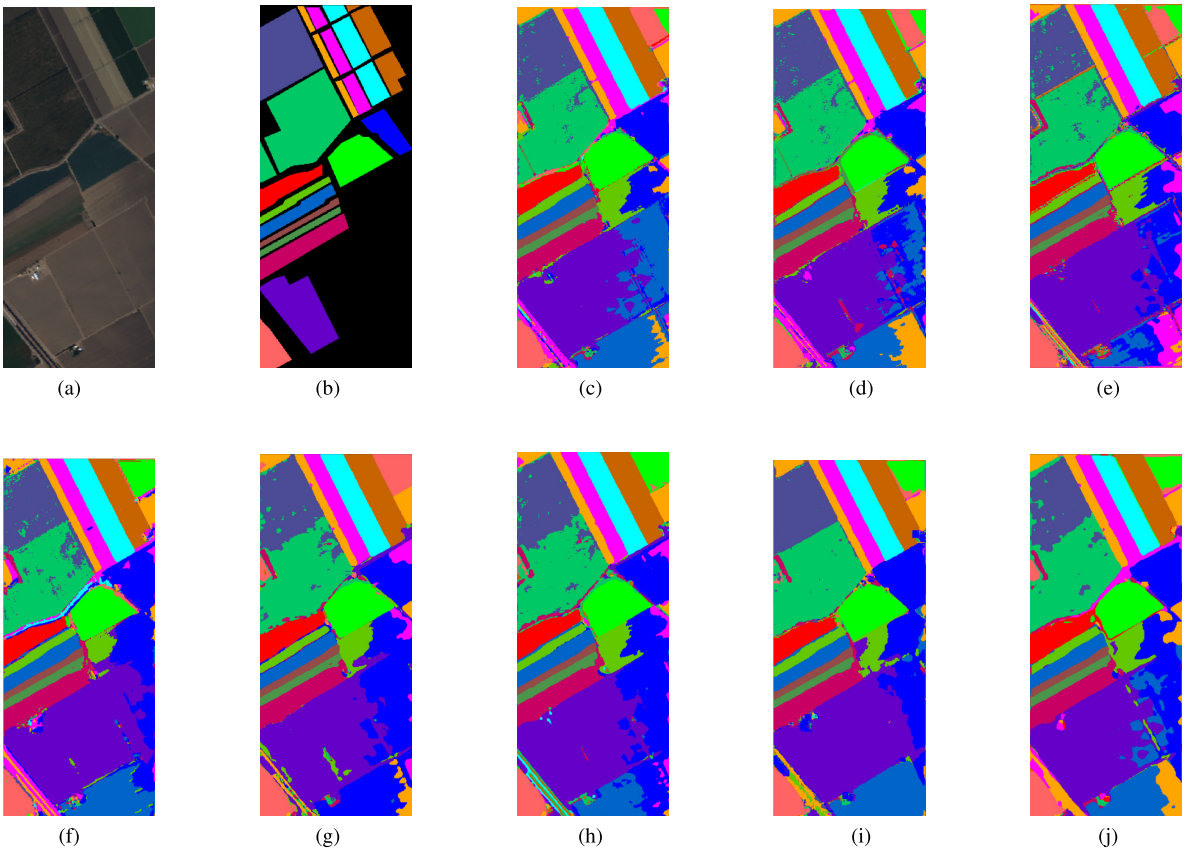

Fig. 8. Classification maps for the SA dataset using 1% of the training samples. (a) False-color image. (b) GT. (c)–(j) Classification maps with disparate algorithms.

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TABLE VIII
CATEGORIZED RESULTS FOR THE PU DATASET USING 1% TRAINING SAMPLES

Class	Color	2D CNNs	1D-2D CNNs	2D-3D CNNs		3D CNNs		Involution	
		PyResNet	CBW	HybridSN	LeeEtAlNet	A2S2KResNet	SSRN	ESSINet-2D	ESSINet-1D
1		94.107	97.477	95.584	75.827	99.076	**100.00**	98.384	98.061
2		98.824	99.042	99.425	92.377	99.929	97.735	**99.940**	99.929
3		65.435	74.526	74.186	40.933	82.596	83.666	**93.680**	94.118
4		93.273	97.269	93.940	93.273	**97.902**	96.170	95.571	94.939
5		99.697	100.00	95.523	**100.00**	99.848	**100.00**	**100.00**	99.848
6		86.224	89.673	90.728	43.092	**99.979**	99.554	99.940	99.838
7		65.004	60.399	80.200	71.681	97.084	**99.693**	97.160	97.084
8		64.579	87.167	85.865	49.351	98.171	**99.390**	97.533	97.143
9		96.121	98.707	96.228	73.922	**99.892**	99.246	97.630	98.707
OA $\times 100$		90.544	94.168	94.325	76.892	98.269	97.805	**98.736**	98.714
AA $\times 100$		84.807	89.362	90.594	70.654	97.181	97.273	**97.760**	97.852
Kappa $\times 100$		87.373	92.227	92.432	68.870	98.003	97.101	**98.322**	98.295
FLOPs		35M	10M	29M	21M	88M	81M	5M	**1M**
Parameters		21.603M	1.597M	1.748M	357.38k	220.82k	216.54k	47.67k	**19.19k**
Inference Time		5.58s	2.08s	6.50s	9.52s	4.61s	3.59s	**1.52s**	1.87s
GPU Memory Usage		1343M	**1185M**	1233M	1235M	1653M	1417M	1217M	1211M

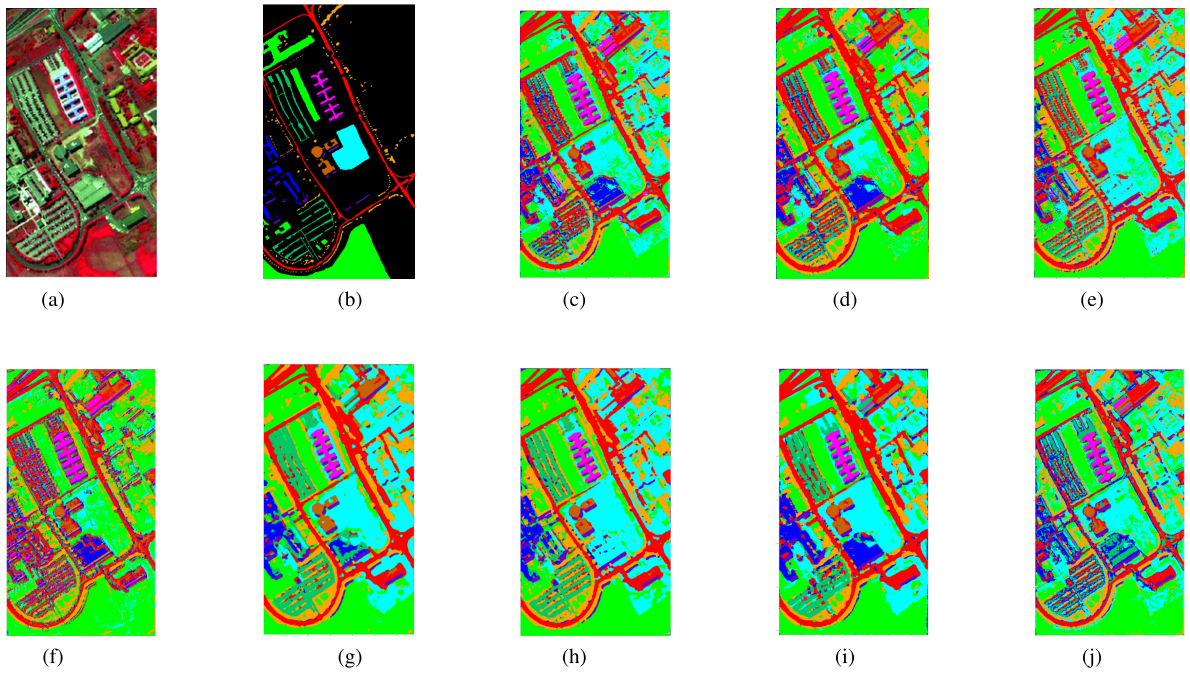

Fig. 9. Classification maps for the PU dataset using 1% of the training samples. (a) False-color image. (b) GT. (c)–(j) Classification maps with different algorithms.

FLOPs, exceeding 100M in three datasets. It is worth mentioning that CBW also reduces a lot of FLOPs due to its 1-D-2-D CNN structure, but its FLOPs are still more than that of ESSINet-1D.

2) On **the number of parameters**, the proposed ESSINet-1D has the fewest number of parameters, and it only needs 19k on the PU dataset. Among the six comparison models, PyResNet has the most number of parameters. Although the FLOPs of CBW are few, it still has lots of parameters because it is still mainly composed of 2-D convolution kernels. Similarly, although the FLOPs of the 3-D CNN models are very high, their number of parameters is smaller than that of 2-D CNN-based models and 2-D-3-D CNN-based models on four datasets.
3) As for **inference time**, taking SA and PU as examples for analysis, it can be seen that the inference time required by ESSINet-2D is the shortest, only 3.03 and 1.52 s, respectively. Note that, although the parameters and FLOPs of ESSINet-1D are the smallest, its inference time is slightly longer than ESSINet-2D. This is mainly because we have adopted some time-consuming reshape operations to achieve the results of 1-D CNN, but we cannot ignore the lightweight level of ESSINet-1D.

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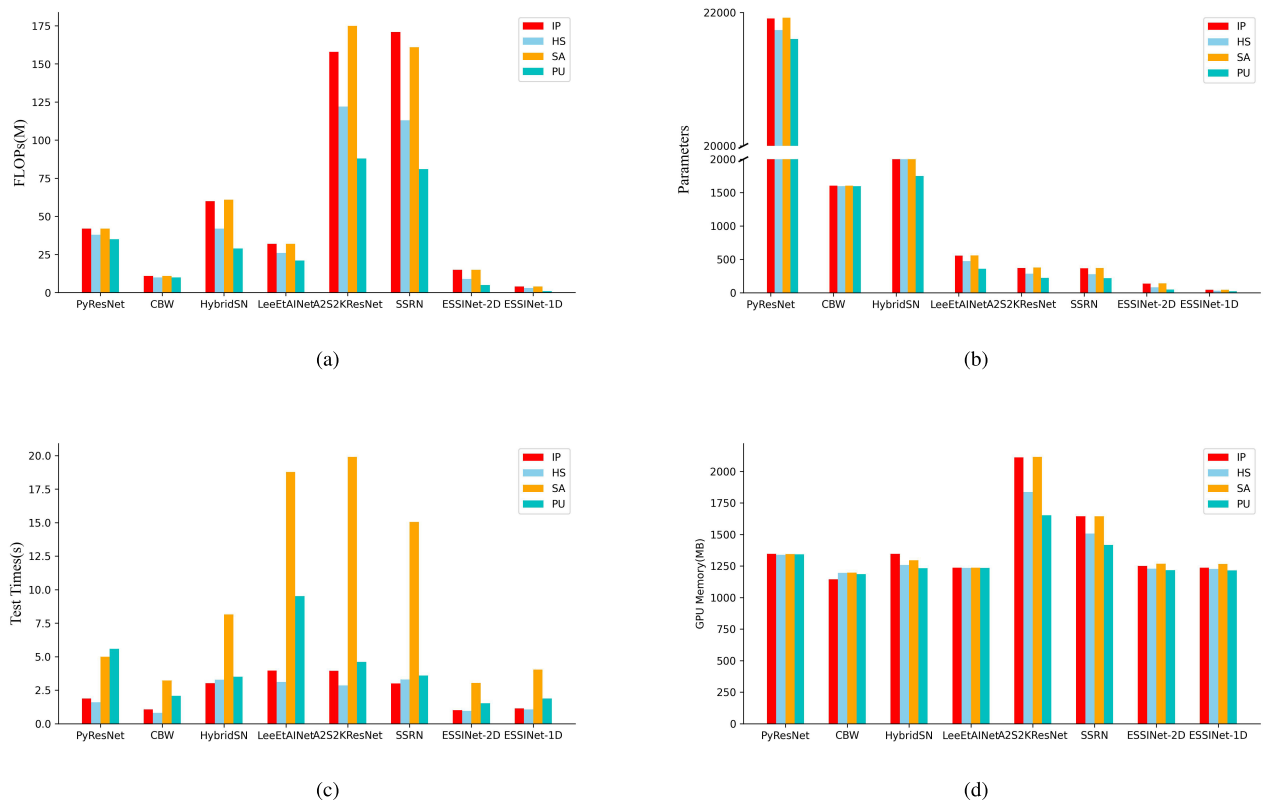

Fig. 10. Efficiency of different models. It shows the comparisons of the number of parameters, FLOPs, inference time, and GPU memory usage for different models under different datasets. (a) Comparison of FLOPs under different datasets. (b) Comparison of model parameters under different datasets. (c) Comparison of inference time under different datasets. (d) Comparison of GPU memory usage under different datasets.

In addition, the inference time of LeeEtAINet is not satisfactory on four datasets.

4) In terms of **GPU memory usage**, the GPU memory usage of CBW is the least in all four datasets, and that of our ESSINets is only slightly more than it, but, considering that our performance, the number of parameters, and FLOPs are better than it, we think that this result is acceptable. As for 3-D CNN-based models, their GPU usage is still large. It can be found that the GPU memory usage of the 3-D CNN models is more sensitive to the dataset.

Generally speaking, the proposed ESSINets show excellent performance in both classification accuracy and model efficiency.

E. Investigation of the Modified Involution and InvBlock

To verify the effects of our proposed involution and the rationality of InvBlock, we perform relevant ablation experiments, and the results are shown in Table IX.

Concretely, for the ESSINet composed of the original involution, its OAs on the IP, HS, SA, and PU datasets are 98.190%, 97.139%, 97.664%, and 97.655%, respectively. Compared with the previous six algorithms, its OA is worse than SSRN and A2S2KResNet in some datasets but better than the other four algorithms, which also confirms that the involution operator has desirable performance. By contrast, the OA of ESSINet-2D is improved to a certain extent, surpassing SSRN and A2S2KResNet, reaching 99.002%, 98.062%, 97.895%, and 98.736%, respectively. In addition, it can be seen that, if any pooling layer is removed, the OA of the model will drop slightly, which also shows the effectiveness of the KGF that we constructed.

TABLE IX
COMPARISONS OF INVOLUTION-2D ON OA

	Ablation	IP	HS	SA	PU
dual pooling	modified involution-2D	**99.002**	**98.062**	**97.895**	**98.736**
	avg pooling removed	98.734	97.799	97.670	97.800
	max pooling removed	98.826	97.623	97.735	97.988
	concat two features	98.981	98.004	97.701	98.411
Invblock	residual connection removed	97.988	96.144	97.309	97.919
	involution-2D replaced	98.121	96.562	96.689	97.064
-	original involution	98.190	97.139	97.664	97.655

We also tried to use sum or concat operation to integrate the two features obtained after the dual-pooling layer. The final experimental result is that the performance of the sum operation is better, and this method can also prevent the increase in the number of parameters and FLOPs.

Table IX also shows the effects of InvBlock on ESSINet-2D. The design of InvBlock is inseparable from the involution and residual connections. When the InvBlock removes the residual connections, the model needs directly learn features, which may increase the difficulty of model training. It can be

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TABLE X
COMPARISON OF DIFFERENT OPERATORS

operator	Parameters	FLOPs	OA (IP)	OA (HS)	OA (SA)	OA (PU)
2D CNN	360k	29M	98.121	96.562	96.689	97.064
original involution	16k	1M	98.190	97.139	97.664	97.655
involution-2D	16k	1M	**99.002**	**98.062**	**97.895**	**98.736**
involution-1D	**4k**	**0.4M**	98.402	97.109	97.089	98.714

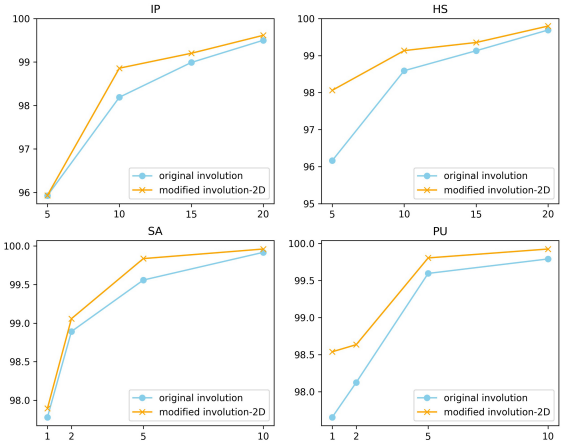

Fig. 11. OA of two methods with varying training set sizes.

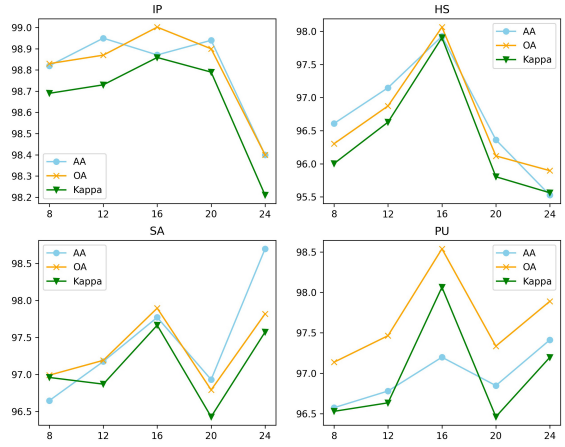

Fig. 12. OA, AA, and Kappa of ESSINet-2D with different values of G on all datasets. $r = 4$ by default.

found that the OA of the models shows varying degrees of decline on the four datasets and is more severe on the IP and HS datasets. When InvBlock replaces the involution-2D with 2-D convolutional kernels, the performance of the model also drops.

To test the effectiveness of the modified involution further, we evaluate the performance of involution on ESSINet-2D before and after modification with varying training sample sizes, i.e., 5%, 10%, 15%, and 20% for IP and HS datasets and 1%, 2%, 5%, and 10% for the PU and SA datasets. Fig. 11 shows the OA of the two methods with varying training set sizes. On the whole, the modified involution-2D consistently outperforms all the datasets. This is mainly because we use modified involution-2D to ensure the space-spectral features more closely interact to extract more discriminative information.

Finally, we make a comparative analysis of the number of parameters and FLOPs between different operators. Taking the input of $9 \times 9 \times 200$ size as an example, the RF is 3, the padding operation is automatically performed, and the results are shown in Table X. It can be found that, compared with a 2-D convolution operator, the involution-2D has a huge decrease in the number of parameters and FLOPs. The added dual-pooling layer enhances the performance of the involution without increasing the number of parameters, and the involution operator is successfully introduced from the computer vision field to the HSI classification task. Moreover, the modified involution-1D based on the involution-2D reduces the number of parameters and FLOPs further, of which the FLOPs are reduced by more than half.

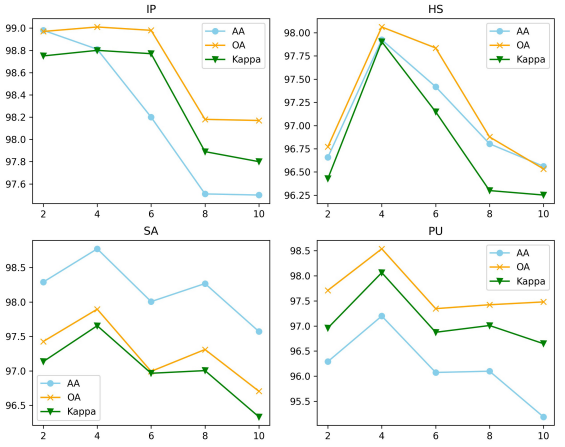

Fig. 13. OA, AA, and Kappa of ESSINet-2D with different values of r on all datasets. $G = 16$ by default.

F. Investigation of the Effects of Different Hyperparameters r and G in ESSINet

For the versatility of the model, ESSINet-2D chooses the combination of $r = 4$ and $G = 16$ as a compromise. However, different r and G will affect the performance of the model. Figs. 12 and 13 show their effect on the proposed ESSINet-2D for OA, AA, and Kappa on the four datasets.

Different r's will first affect the KGF and then affect the feature extraction ability of the involution kernel. The smaller the r value, the more features that can be found by the model, but it will lead to a larger number of parameters and FLOPs in

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KGF. A different G value will affect the number of channels in the involution kernel. As the G value increases, the number of channels in the involution kernel increases, which means that involution divides the feature map channels into more groups. The larger the G value, the greater the number of parameters and FLOPs.

It can be found that, when $G = 16$ and $r = 4$, ESSINet-2D performs desirably on each dataset. However, choosing an appropriate r and G can make the model achieve better results on different datasets. For example, when $G = 24$ and $r = 4$, the AA obtained by ESSINet-2D in the SA dataset is the best, which shows that each category of the SA dataset has a better predictive ability.

V. Conclusion

In this article, we design two novel involution operators based on the characteristics of the HSI, called involution-2D and involution-1D. These operators add a dual-pooling layer on the basis of the original involution so that it can also utilize the features of the spatial domain. The difference between the two modified involutions is what the convolution behind the dual-pooling layer is. Then, we propose a new supervised HSI classification network ESSINet, which has different properties and characteristics from the CNN-based methods. This algorithm is common to both involution-2D and involution-1D. ESSINet-2D using involution-2D achieves the best performance on four public datasets compared with the other six existing algorithms. While among the eight algorithms, ESSINet-1D using involution-1D has the least number of parameters and FLOPs, and it also has comparable performance to some 3-D CNN-based models.

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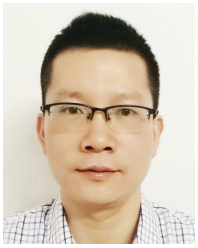

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